

Big Data Analytics
CISELEC 501A-CS41S1

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Laboratory Activity #07: Predictive Analysis
Total Points: 100

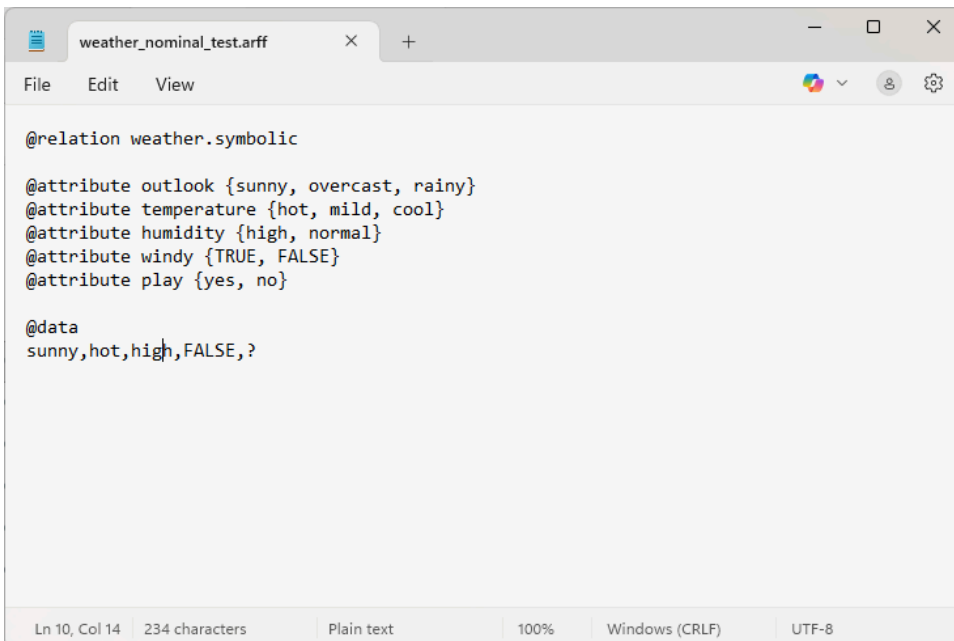
I. OBJECTIVE:

- A. The objective of this laboratory exercise is to enable students to understand and apply predictive analysis using WEKA. Through this activity, students will learn that predictive modeling is achievable using any of the classification algorithms available in the software. By utilizing a given training dataset, the chosen algorithm will identify patterns and relationships within the data, which will then be applied to the testing dataset to generate predictions. This exercise aims to provide students with practical experience in executing the fundamental steps of predictive analysis, including model training, testing, and evaluation. Furthermore, it seeks to deepen students' comprehension of how classification algorithms can be used to make data-driven forecasts and support informed decision-making in various analytical contexts.

II. PROCEDURE

- A. 1.) Create a folder and download the “weather_nominal_training.arA” dataset file from the Git repository inside the “20251108-Lab-Wk14” directory.
- B. 2.) In your local directory, make another copy of the “weather_nominal_training.arA” file and rename it to “weather_nominal_test.arA”.
- C. 3.) Open the newly created file using a text editor. Remove or delete lines 11 and onwards; retain only the following contents:
- D. 4.) Edit the last line to:
- E. 5.) The contents of “weather_nominal_test.arA” file should to the example below:
- F. 6.) Open your WEKA and launch the “WEKA Explorer” window. In the Preprocess tab, open the “weather_nominal_training.arA” data set.
- G. 7.) Go to “Classify” tab and follow the configurations below:
- H. 8.) Once completed, enable the “Supplied test set” radio button under the “Test options” section and press the “Set...” button to open the “Test instances” modal window. Load your “weather_nominal_test.arA” file by pressing the “Open file...” button and locate the file. Press the “Close” button once the test file is loaded.
- I. 9.) Press the “More options...” and follow the configurations below:
- J. 10.) Press the “OK” button once the setup is complete.
- K. 11.) Press the “Start” button to execute the classification process.

L. IMAGES:

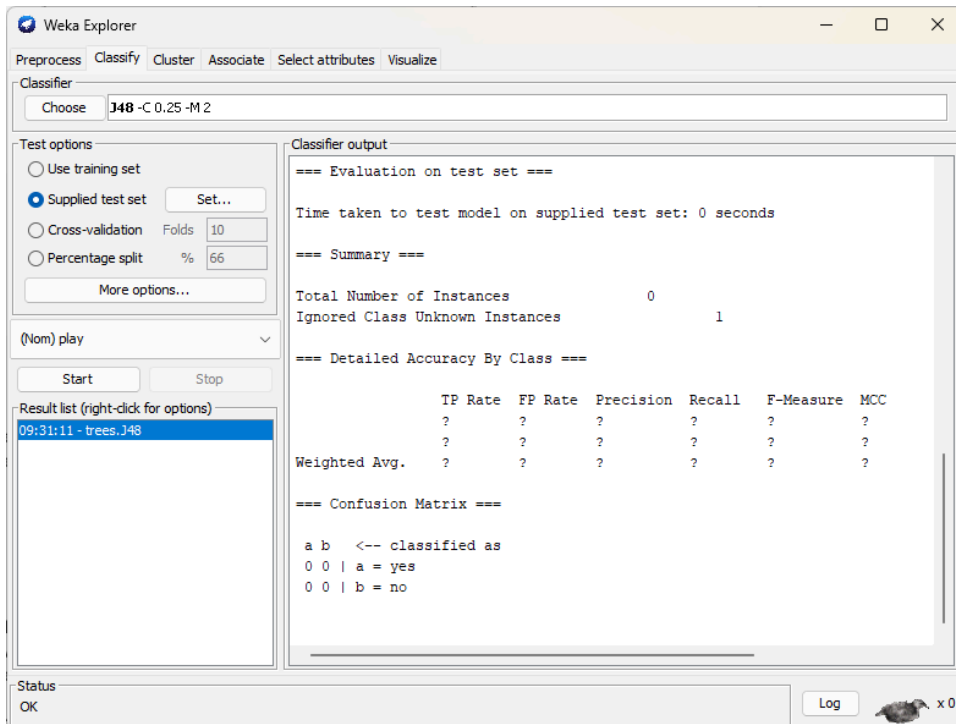
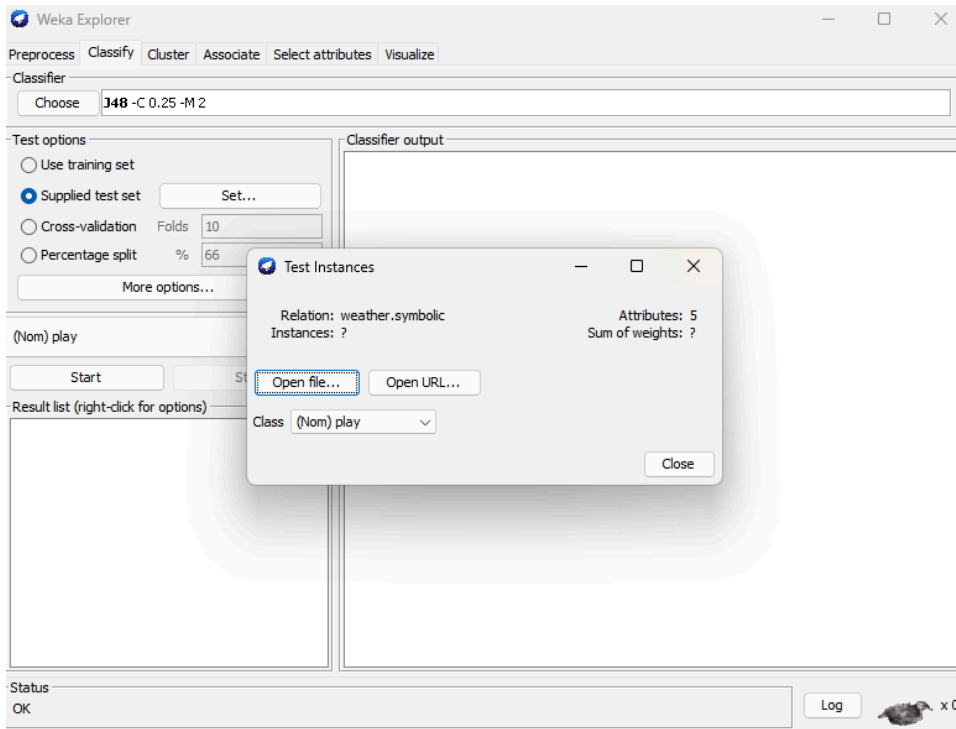


```
@relation weather.symbolic

@attribute outlook {sunny, overcast, rainy}
@attribute temperature {hot, mild, cool}
@attribute humidity {high, normal}
@attribute windy {TRUE, FALSE}
@attribute play {yes, no}

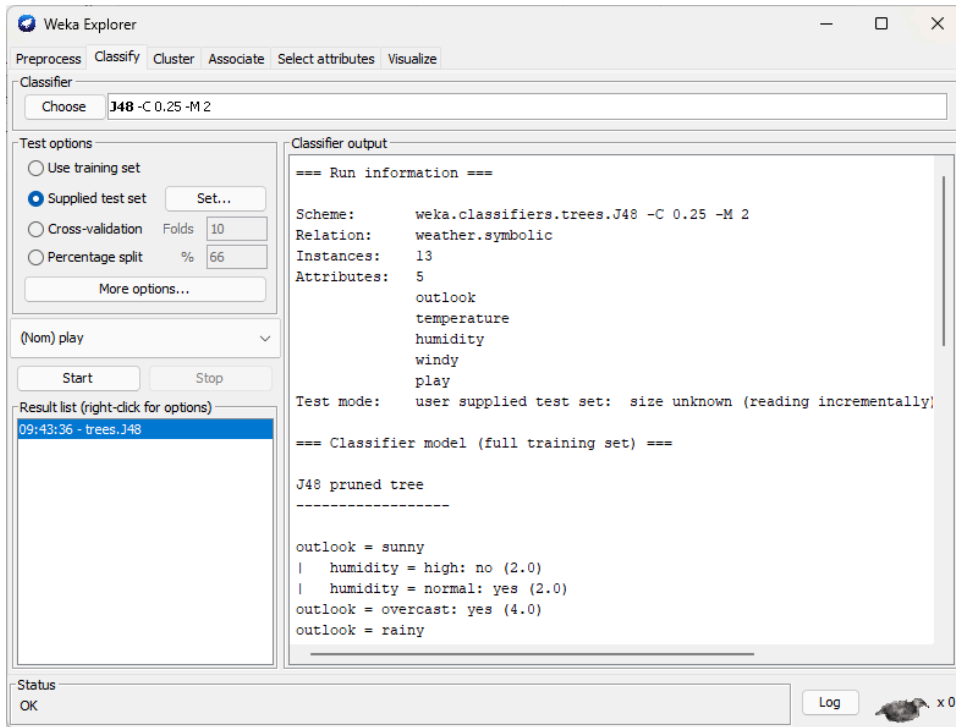
@data
sunny,hot,high,FALSE,?
```

Ln 10, Col 14 | 234 characters | Plain text | 100% | Windows (CRLF) | UTF-8



III. REFLECTION (50 pts)

- A. 1) Under the “Classifier output” section, locate “=== Predictions on test set ===” area and check the value under “Prediction” column. What was the output provided by the J48 algorithm? (5 pts)
 1. The prediction column only show the index of the predicted class which is ‘1’ but to see the actual value of the prediction, we can look at the **predicted** column and see that it is ‘no’
- B. 2) What will be the output of the entire process if the line “sunny,hot,high,FALSE,no” is removed from the training data set? (15 pts)



1. The output is still the same. Our training dataset predictively outputted the same output with or without the object "sunny,hot,high,FALSE,no"
- C. 3) Select two (2) other classification algorithms, rerun the same procedure, and record their "Prediction" results. **(30 pts)**
 - 1.

Classifier	Prediction Result
trees.J48	1 (no)
bayes.NaiveBayes	0.704(no)
lazy.IBk	0.938(no)

IV. ADDITIONAL TASK (50 pts)

- A. Follow the same procedure as demonstrated in the previous activity, and search for a suitable dataset available on Kaggle.com. Ensure that the dataset you selected contains a clear identified class or categorical attributes for each record, as this will be essential for performing classification and predictive analysis using the chosen algorithm. Record your results using the table provided in section III (Reflection), item #3.
- B. KAGGLE SOURCE:
 1. [Heart Disease](#)
 2. target class: **Heart_Disease_Status**
 3. Algorithm Chosen: **RandomForest**
 4. Handling Imbalance: **Resample Filter**
 5. Training: **Cross-Validation 10 Folds**

C. SCREENSHOTS

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose

RandomForest -P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1

Test options

Use training set

Supplied test set

Cross-validation

Folds

10

Percentage split

%

66

More options...

(Nom) Heart_Disease_Status

Start

Stop

Result list (right-click for options)

11:29:29 - trees.RandomForest

11:30:00 - trees.J48

11:30:52 - trees.RandomTree

11:33:25 - trees.RandomTree

11:33:33 - trees.RandomForest

11:34:00 - trees.J48

11:36:09 - trees.RandomForest

11:36:21 - trees.RandomTree

11:36:39 - trees.J48

11:37:13 - trees.RandomForest

Classifier output

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	9308	93.0986 %
Incorrectly Classified Instances	690	6.9014 %
Kappa statistic	0.862	
Mean absolute error	0.2115	
Root mean squared error	0.2766	
Relative absolute error	42.3017 %	
Root relative squared error	55.3238 %	
Total Number of Instances	9998	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC
	0.962	0.100	0.906	0.962	0.933	0.864
	0.900	0.038	0.959	0.900	0.929	0.864
Weighted Avg.	0.931	0.069	0.933	0.931	0.931	0.864

=== Confusion Matrix ===

a	b	<-- classified as
4807	192	a = No

Status OK Log x 0

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose

RandomForest -P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1

Test options

Use training set

Supplied test set

Cross-validation

Folds

10

Percentage split

%

66

More options...

(Nom) Heart_Disease_Status

Start

Stop

Result list (right-click for options)

11:43:47 - trees.RandomForest

Classifier output

Time taken to build model: 2.32 seconds

=== Predictions on test set ===

inst#	actual	predicted	error	prediction
1	1:?	1:No	0.946	
2	1:No	1:No	0.927	
3	1:?	1:No	0.93	
4	1:No	1:No	0.88	
5	1:No	1:No	0.87	
6	1:?	1:No	0.88	
7	1:?	1:No	0.97	
8	1:?	2:Yes	0.73	

=== Evaluation on test set ===

Time taken to test model on supplied test set: 0 seconds

=== Summary ===

Correctly Classified Instances	3	100 %
Incorrectly Classified Instances	0	0 %

Status OK Log x 0

Classifier	Actual	Prediction
trees.RandomForest	?	0.946(no)
trees.RandomForest	no	0.927(no)
trees.RandomForest	?	0.93(no)
trees.RandomForest	no	0.88(no)
trees.RandomForest	no	0.87(no)
trees.RandomForest	?	0.88(no)
trees.RandomForest	?	0.97(no)

Classifier	Actual	Prediction
trees.RandomForest	?(but actually yes)	0.73(yes)

" I affirm that I have not given or received any unauthorized help in
this assignment, and that this work is my own "

Kenneth Arias

A handwritten signature in black ink, appearing to read 'Kenneth Arias', with a stylized, cursive script.